



A: Description

Marks: 80% of Marks will be allocated.

Duration: 14 Weeks.

Assignment: As announced during the first lecture, and well briefed during subsequent lectures, Q&A Sessions, this is a project spans through the whole semester. The final outcome written appropriately on A4 papers on both sides and should be submitted by the end of the ongoing July 2026 semester.

B: Background

1 Objective

The objective of this assignment is to demonstrate the practical application of Machine Learning concepts learned during the course. Students are required to select a simple (naïve) dataset, apply one or more machine learning models covered during lectures, evaluate model performance, and present their work in the format of an IEEE conference paper (The template is supplied herewith).

1.1 Machine Learning Models

Students must use only the machine learning algorithms covered during lectures.

1.2 Data Preparation

Perform necessary preprocessing activities.

1.3 Performance Evaluation

1.3.1 Classification Problems

Include the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score

- Confusion Matrix
- ROC Curve and AUC

1.3.2 Regression Problems

Include:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

1.3.3 Clustering Problems

Include:

- Silhouette Score
- Davies-Bouldin Index
- Cluster Visualizations

1.4 Results Analysis

Discuss:

- Model performance
- Comparison of multiple models (only if applicable)
- Strengths and limitations
- Interpretation of results
- Key findings

2 Final Deliverables

2.1 IEEE Conference Paper

Prepare the report using the IEEE Conference Paper format. Use IEEE citation style. Recommended length: 4–6 pages maximum (including references).

2.2 Source Code

You may use the codes supplied within the Jupyter Notebooks provided for each course sub-module with proper citations. Submit:

- Notebook (.ipynb) or Python Script (.py) format.
- **Well-commented** source code.
- Execution instructions.

2.3 Plagiarism Report (Mandatory)

Students must submit a plagiarism/similarity report generated using an approved plagiarism detection tool (e.g., Turnitin, or institutional software). Requirements:

- The plagiarism report must be attached to the final submission.
- The submitted paper must comply with the institutional plagiarism policy.
- Similarity percentage should not exceed the limit specified by the department.
- Failure to submit the plagiarism report of exceeding the specified limit result in penalties or rejection of the submission.

3 Submission Requirements

Table 1: Required Submission Items

Item	Format
IEEE Conference Paper	PDF
Source Code	.ipynb or .py
Plagiarism Report	PDF
Dataset (Optional)	ZIP
Supporting Files (Optional)	ZIP

4 Assessment Criteria

5 Important Notes

1. This assignment is to be completed individually unless otherwise instructed.
2. Use only machine learning algorithms covered in lectures.
3. Clearly distinguish training, validation, and testing datasets.

Table 2: Marking Scheme

Criterion	Weight (%)
Understanding the Dataset	05
Data Preprocessing and EDA	05
Application of the Model(s)	10
Training, Testing and Validation	05
Performance Evaluation	10
Interpretation, Analysis of Results	40
Discussion	20
IEEE Paper Quality and Presentation	05
Total	100%

4. All figures and tables must be properly labeled and referenced.
5. All already published information or data must be properly cited and referenced.
6. Academic integrity policies must be strictly followed.
7. **Plagiarism and unauthorized use of AI-generated content without proper acknowledgment may result severe penalties.**
8. Attach the plagiarism report together with the final paper submission.

Submission Deadline

As announced at the commencement of the semester, on or before the last date of the ongoing July 2026 semester.